2	**(a)**		obeys the law pV/T = constant <u>or</u> any <u>two</u> named gas laws	**M1**	
			at all values of p, V and T	**A1**	**[2]**
			<u>or</u> two correct assumptions of kinetic theory of ideal gas **(B1)**		
			third correct assumption **(B1)**		
	(b)	**(i)**	mean square speed	**B1**	**[1]**
		(ii)	mean kinetic energy = $\frac{1}{2}m\langle c^2\rangle$	**M1**	
			$\rho = Nm/V$ and algebra leading to [do not allow if takes N = 1]	**M1**	
			$\frac{1}{2}m\langle c^2\rangle = 3/2\ kT$	**A0**	**[2]**
	(c)	**(i)**	$\frac{1}{2} \times 6.6 \times 10^{-27} \times (1.1 \times 10^4)^2 = 3/2 \times 1.38 \times 10^{-23} \times T$	**C1**	
			$T = 1.9 \times 10^4$ K	**A1**	**[2]**
		(ii)	Not all atoms have same speed/kinetic energy	**B1**	**[1]**